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MILITARY SPECIFICATION

TRACK SHIFTERS, RAILWAY, GASOLINE- OR DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power-driven track shifters for alining track in railway maintenance and rehabilitation operations.

1.2 Classification. Track shifters shall be of the following types or classes as specified (see 6.3):

Type I - 36-inch (914 mm) gage.
Type II - 39-3/8-inch (1 000 mm) gage.
Type III - 42-inch (1 067 mm) gage.
Type IV - 56-1/2-inch (1 435 mm) gage.
Class I - Gasoline-engine-driven.
Class II - Diesel-engine-driven.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 2230

Beneficial comments (recommendat6ons, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation-Packaging, Method of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Heavy Duty.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-M-18058	- Machinery, Metal and Wood-Working, Preparation for Delivery of.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1410	- Method for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS35832	- Diagram, Equipment, Composite, Railway, (36, 39-3/8 and 42 inch) Gages, Foreign Service.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

J512 - Automotive Tube Fittings.
J514 - Hydraulic Tube Fittings.
J516 - Hydraulic Tube Fittings.
J517 - Hydraulic Hose.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. Railway track shifters shall be portable gasoline- or diesel-engine-driven, hydraulic-powered unit for alining railroad track having 36-, 39-3/8-, 42-, or 56-1/2-inch (914-, 1 000-, 1 067-, or 1 435-mm) gage. The frame shall be constructed of steel. Each track shifter shall require only one man operation and be capable of alining track weighing from 45 to 155 pounds per yard (22 to 77 kg per m). Track shifters shall be capable of controlling the throw of track horizontally from 1/32- to 6-inches (1- to 152-mm) and simultaneously alining track at a rate of at least 1,000 feet (300 m) per hour.

3.2 First article (first-produced track shifter). The contractor shall furnish a track shifter for examination and testing within the time frame specified to prove that his production methods and choice of design detail will produce track shifters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Threaded parts. All threaded parts shall be in accordance with Handbook H28.

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3.4 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.5 Human factors engineering. All controls required for the operation of the shifter shall be within reach of the operator seated in the operator's seat. The operator shall have a clear view of the track (to front and rear) and of the operation of the shifting mechanism and jacks. The noise level at the operator's station shall not exceed the limits specified in MIL-STD-1474, Catetory C for steady state noise when tested in accordance with 4.5.2.11 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.11. Hearing protectors shall not be supplied with the tamper. Hazard sign, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED".

3.6 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, or replacement. Covers or plates must be readily and easily removed for component adjustment, replacement, or maintenance.

3.6.1 Filling, draining, and checking provisions. Enclosures such as hydraulic system reservoirs, gearcases and transmission housings that contain a reservoir of oils and lubricants shall be equipped with dipsticks, check plugs or sight gages to determine the level of the fluid. Each enclosure shall be equipped with a means for filling the enclosure with fluid and for draining. The drain outlet shall be located so that removal of the plug will result in complete drainage of the fluid from the enclosure. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.7 Overall dimensions and weight. The dimensions and weight of each track shifter in travel position shall be as follows:

Weight with setoff	10,000 pounds (4 536 kg)
Length	180 inches (4 572 mm)
Width	90 inches (2 286 mm)
Height	125 inches (3 200 mm)

The width and height of Types I, II, and III shall conform to MS35832.

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3.8 Hydraulic system. The hydraulic system on the vehicle shall provide a means of operating the track shifter. Oil for the system shall conform to MIL-L-2104, Grade 10. For low temperature operation, oil conforming to MIL-L-46167 may be used. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage, or permanent deformation. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the vehicle.

3.8.1 Test points. One test point shall be located at the outlet of each pump for measuring pressure.

3.8.2 Cylinders. All hydraulic cylinders shall be double-acting, and shall be fabricated from seamless steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device at the rod to prevent foreign matter and fluids from entering and damaging the seal and other cylinder components. All cylinders shall perform as specified herein without buckling or bending. The cylinders shall be located as close as practicable to the track wheels and shall operate from hydraulic pressure supplied by pump(s) powered by the engine. The piston rod shall be hard chromed plated to prevent corrosion. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder.

3.8.3 Pumps. The track shifter shall be equipped with hydraulic pump(s) driven directly by a gasoline or diesel engine. The pump(s) shall be of a constant displacement type and shall be capable of supplying the system relief valve setting pressure at a speed not exceeding those of the manufacturer's rated speed.

3.8.4 Controls. Cylinders shall be provided with individual controls. All hydraulic control valves shall be of the metering type so that accurate control of the unit may be maintained at all times. All controls shall be located within easy reach of the operator. Pushing of a lever shall result in a lowering or extending motion, and pulling on the lever shall result in a raising or retracting motion. Forces necessary to actuate the controls shall be 10 pounds (5 kg) maximum. All control valve spools shall be spring-centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals. Spool shall not bind under all operating conditions specified herein. Hydraulic control valves shall unload the pump when cylinders or propelling motor are not in use.

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3.8.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be obtained only when a cylinder is bottomed or when the system is subjected to shock loading.

3.8.6 Return line filter. The system filter element(s) shall conform to MIL-F-52723. The filter shall maintain a full flow condition during all operations of the jack. The filter bypass may open for system oil at a temperature less than 70° F (21° C). All return oil shall pass through the filter except relief oil which may return directly to the reservoir. An automatic bypass that permits full flow at a pressure of not less than 35 psid (241 kPa) shall be provided in the filter housing. The filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall show when the element is clean and, as a minimum, when the filter is beginning to bypass, requiring a change of the element.

3.8.7 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that may result in leaks, loosening of fasteners or cracks. Connections on the reservoir shall be located so that connecting lines can be removed without removing the reservoir or any other major component. The reservoir shall be located and constructed so that during operation and performance of all functions specified herein the pump shall not be subject to aeration. The reservoir should be baffled when necessary to provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. A dipstick or sight gage shall be furnished to determine the reservoir oil level. The dipstick shall be marked "Add" and "Full" by etching, stamping, engraving or embossing. The dipstick shall be removable without tools and shall be located in the filler neck. Access to the interior of the reservoir shall be provided for repair and removal of all internal parts and components. The filler cap and tube shall be located to prevent entrance of contaminants throughout filling and checking operations. The reservoir pump suction openings should be located a minimum of 1 inch (25 mm) above the bottom of the reservoir to prevent entrance of settled contaminants. Except for sealed reservoir, a cover breather shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain.

3.8.8 Hydraulic lines. Hydraulic lines shall consist of tubing or pipe, hose, and fittings. Hose or a combination of hose and tubing shall be used between components that move in relation to each other. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance and maximum protection. Provision shall be made to prevent

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damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines other than suction lines shall be considered pressure lines. All lines shall be cleaned free of foreign matter including weld slag and spatter.

3.8.8.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 and pressure hose couplings shall be as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum working pressure (relief valve setting). The hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.8.8.2 Pressure tubing and piping. Pressure tubing and piping shall have a minimum burst pressure rating of 3.5 times the system maximum working pressure (relief valve setting). All tubing bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows: Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter. All tubing and piping shall be cleaned free of foreign matter including cuttings and weld slag and spatter.

3.8.8.3 Pressure pipe fitting and pressure tube fittings. Unless otherwise specified herein, pressure tube fitting shall conform to either SAE J512 or J514. Pressure tube fitting shall have burst pressure rating that is equal to or greater than 3.5 times the maximum system working pressure (relief valve setting).

3.9 Lining mechanism. The track shifter shall be provided with spud, shifting dogs, and alining head, controlled by the operator. The spud and shifting dogs shall be mounted on the alining head.

3.9.1 Alining head. The alining head shall be hydraulically raised and lowered. When lowered into the crib, the alining head shall position the spud and shifting dogs for the track alining operation. In raised position, no part of the alining head, spud, or shifting dogs shall be less than 3 inches (76 mm) above the top of the rail.

3.9.2 Spud. The track shifter shall be firmly anchored by lowering the spud into the roadbed with the alining head lowered into the crib. The spud shall be powered by double-acting hydraulic piston(s) or by a hydraulic motor through a gear reducer. Each track shifter shall be provided with a mechanical lock to relieve spud pressure and to prevent dropping the spud while in transit or storage.

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3.9.3 Shifting dogs. Each track shifter shall be equipped with shifting dogs to push against the base of the rails. Each dog shall have a cylinder for actuation and be provided with controls to regulate the amount of pressure applied against the rail.

3.10 Bearings. All bearings (except engine bearings) shall be anti-friction bearings of the ball, roller, or tapered roller type.

3.11 Wheels. Each track shifter shall be mounted on four ductile-iron, cast-steel, or pressed-steel, single-flanged wheels not less than 16 inches (406 mm) in diameter for movement on track.

3.12 Braking system. The track shifters shall be equipped with a hydraulically controlled or mechanical braking system that will hold the track shifter in a stationary position when on a 5 percent grade.

3.13 Turntable. Each track shifter shall be furnished with a built-in, center-lift hydraulic turntable, which shall rotate freely 360 degrees. The track shifter shall have facilities for storing the turntable when not in use. Not more than three men shall be required to remove the track shifter from the rails and place it to the side and clear of the track. Removal of the track shifter from and clear of the track and reversing the operation by setting the track shifter back on the railroad track shall be completed in not more than 10 minutes.

3.14 Setoff rails. Setoff rails with adjustable rail supports shall be furnished and stored on each track shifter.

3.15 Power transmission. Engine power shall be transferred to the wheels by mechanical or hydraulic transmission. The transmission shall have at least two speeds in both the forward and reverse directions. The track shifter speed on level track and in zero wind conditions shall be between 4 and 8 mph (6 and 13 kmph) in the lowest gear and at least 20 mph (32 kmph) in the highest gear.

3.15.1 Clutch. When a clutch is provided, it shall be located between the engine and the transmission. The clutch shall be rated by its manub facturer at not less than 150 percent of the maximum torque developed by the engine.

3.16 Engine. For the purpose of selecting the engine, the track shifters shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine, including all systems, components and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. The engine for the Class I track shifters (see 1.2), shall be gasoline and the engine for the

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Class II track shifters (see 1.2), shall be diesel, unless otherwise specified (see 6.2). The maximum governed speed for each engine, as applied, shall not exceed the engine manufacturer's recommended speed for this application. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall meet all operational requirements and conditions specified herein without exceeding the limits specified in Table I of MIL-STD-1410, or, the engine manufacturer's recommended operating limits. When a diesel engine is furnished, the exhaust smoke shall not exceed 3.0 when tested as specified herein. The average oil consumption of each engine shall not exceed 3.0 when tested as specified herein.

3.16.1 Fuel tank. When either type engine is specified, a fuel tank shall be furnished with a capacity sufficient for 8 hours of continuous operation in any operating condition specified herein.

3.16.2 Battery and battery box. Unless otherwise specified (see 6.2), the tamper shall be furnished with batteries conforming to SAE J537H. A weatherproof box made of steel not less than 0.109 inch (3 mm) thick shall be provided for the batteries. With the batteries installed in the box, with all electrical connections completed, and with the battery-box lid in the closed position, there shall be no noninsulated electrical conductor within 1 inch (25 mm) of any point within the battery box where an electrical potential exists. Vents shall be provided near the top of the box, and drains shall be provided in the bottom. The box shall have built-in restraints and holddowns constructed so that pressure will be applied only to the outer battery case. The box shall be located so that the batteries will be accessible for routine maintenance, removal, or replacement. It shall be possible to service the batteries using conventional, battery-servicing aids such as a hydrometer and a syringe. Grommets shall be furnished to protect the battery-lead cables. All interior surfaces of the box shall be treated with an acid-resistant paint.

3.16.3 Accessories. The engine shall be furnished with the following accessories: A housing; an electrical-cranking system; a battery-charging system; and an instrument panel equipped with but not limited to the following instruments, as applicable:

(a) Gage, lubricating oil.
(b) Indicator, fuel, liquid quantity.
(c) Indicator, cooling-liquid-temperature.
(d) Indicator, battery charging.

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(e) Tachometer.
(f) Hourmeter.

The panel shall be located and mounted in clear view of the operator seated in the operator's seat.

3.17 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.17.1 Lubricants. The track shifters including hydraulic systems, shall operate as specified herein when lubricated with military lubricants (see 6.5), or as the manufacturer recommends.

3.17.2 Lubrication fittings. Lubrication fittings, when used, shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun and a 10-inch (254-mm) flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings.

3.17.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.17.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be in accordance with commercial practice for this type of equipment.

3.17.5 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.18 Lifting provisions. Permanent lifting provisions that enable the track shifter to be lifted in its normal travel position shall be provided.

3.19 Tiedown provisions. Tiedown provisions that permit fastening the track shifter to the floor or deck of a transportation medium shall be provided. Tiedown provisions may also be used as lifting provisions when such attachments meet the requirements specified in 3.19.

3.20 Identification marking. The track shifter shall be identified in accordance with MIL-STD-130. The marking shall be applied to the track shifter on plates conforming to MIL-P-514, Type I, Style 1, Composition C.

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3.21 Shipping data plate. Shipping data plate shall conform to MIL-P-514, Type III, Composition C material and in addition, shall show the silhouette of the track shifter in transport position indicating the center of gravity and the location and capacity of the lifting and the tiedown provisions. The plates shall be attached to the track shifter.

3.22 Instruction plates. Each track shifter shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the track shifter. Instruction plates shall conform to MIL-P-514, Type III, Composition C. Plates shall be attached in a conspicuous protected location.

3.23 Treatment and painting. The portions of the track shifter normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, color yellow.

3.24 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.25 Items to be furnished with track shifters. No shipment of track shifters shall be made unless repair parts, maintenance tools, and accessories are included with the shipment or unless approval for shipment of the track shifters without such items has been received from the contracting officer.

3.26 Toolbox. Each track shifter shall be provided with a weatherproof toolbox made of steel. The toolbox lid shall open to at least 90 degrees from the closed position. The lid shall remain in the fully open position either by its own weight or by holding devices. Fasteners shall be provided to keep the lid closed. A hasp which will accommodate a keylock shall be provided for locking the toolbox. The toolbox shall be fastened to the track shifter in a protected position. A drain hole shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the maintenance tools and repair parts specified in 3.25.

3.27 Workmanship. All parts, components, and assemblies of the track shifter including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except when sharp edges and corners are required.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced track shifter inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First-produced track shifter inspection.

4.3.1 Examination. The first-produced track shifter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced track shifter shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each track shifter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each track shifter shall be tested as specified in 4.5.2.1 and 4.5.2.3. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The track shifters shall be examined for the following defects:

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101. Dimensions and weight not as specified.
102. Threaded parts not as specified.
103. Safety appliance not as specified.
104. Human factors engineering not as specified.
105. Fluid drains not accessible for draining.
106. Covers and enclosures not equipped with quick-disconnect fasteners for maintenance.
107. Fluid filling, draining, and checking provisions not as specified.
108. Hydraulic pumps or pump drives not as specified.
109. Hydraulic-system filters not as specified.
110. Hydraulic hose and tubing not as specified.
111. Hydraulic cylinders not as specified.
112. Hydraulic valves not as specified.
113. Track shifting mechanism not as specified.
114. Alining head clearance not as specified.
115. Spud not equipped with mechanical lock.
116. Bearings not as specified.
117. Wheels not as specified.
118. Adjustable setoff rails not supplied.
119. Engine not as specified.
120. Battery enclosure not as specified.
121. Engine accessories missing.
122. Surfaces requiring lubrication not provided with a means for lubricating.
123. Pressure-release device not as specified.
124. Identification, shipping data, or instruction plates not as specified.
125. Treatment and painting not as specified.
126. Components missing or not as specified.
127. Grease oil tags missing.
128. Repair parts or maintenance tools not furnished when specified.
129. Toolbox not as specified.
130. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the track shifter shall be lubricated with oils and greases specified in 3.18.1. Oils shall be those designated for use in the ambient temperature at the place of test. During all tests, the engine shall be governed at a speed not greater than the continuous rated speed as established by the engine manufacturer. Hydraulic pumps shall not exceed the continuous speed and pressure recommended by the pump manufacturer during the tests. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test.

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4.5.2.2 Hydraulic system. The hydraulic components, except reservoir, shall be pressurized to not less than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) required to meet the requirements of this specification. The reservoir shall be pressurized to 3.5 psi. Any external leaks, deformation, or breakage of any component shall constitute failure of this test.

4.5.2.3 Operational test. Operate each track shifter for not less than 15 minutes manipulating all controls. Any operational difficulties, breakage or permanent deformation shall constitute failure of this test.

4.5.2.4 Travel. Place the track shifter on a level section of railroad track. One man shall operate the track shifter. Operate the track shifter between 4 and 8 mph (6 and 13 kmph) in low gear and not less than 20 mph (32 kmph) in high gear in both forward and reverse directions. Inability to operate between 4 and 8 mph (6 and 13 kmph) in low gear in forward and reverse or at least 20 mph (32 kmph) in high gear in forward and reverse shall constitute failure of this test.

4.5.2.5 Brake. Place the track shifter on a 5 percent grade and set the brake. Failure to remain stationary with the brake set shall constitute failure of this test.

4.5.2.6 Performance. Operate the track shifter on rail weighing 155 pounds (70 kg) per yard. Move the track shifter horizontally including throw of 1/32 inch (1 mm) and 6 inch (152 mm) for 1/4 mile (0.4 km) and simultaneously aline the track at a rate of not less than 1,000 feet (300 m) per hour. Any operational difficulties, breakage, evidence of contamination on hydraulic filter, leakage, permanent deformation, inability of the track shifter to travel, operate, and move track 1/32 inch (1 mm) and 6 inches (152 mm) for 1/4 mile (0.4 km) or aline the track at a rate of not less than 1,000 feet (300 m) per hour shall constitute failure of this test.

4.5.2.7 Turntable. Raise the track shifter on the turntable and rotate the track shifter at least 360 degrees. Failure of the track shifter to rotate 360 degrees on the turntable shall constitute failure of this test.

4.5.2.8 Portability. Remove the track shifter from the railroad track onto the setoff rails and move to the side and clear of the track using only three men and the turntable. Then reverse the operation and set the track shifter back on the railroad track. Inability to complete the test in 10 minutes shall constitute failure of this test.

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4.5.2.9 Lifting and tiedown provisions. Apply the vectorial force specified in 3.20 for lifting provisions and the static load specified in 3.21 for tiedown provisions to each provision when mounted on its structural member. Weld failure or permanent deformation of any attachment, fastening, or structural member shall constitute failure of this test.

4.5.2.10 Noise level measurement. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady state noise while operating the jack. Nonconformance to 3.5 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking shall be examined and tested to determine compliance with the quality assurance provisions of the referenced documents in Section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C as specified (see 6.2).

5.1.1 Level A. The track shifter with accessories, repair parts, and maintenance tools, shall be preserved in accordance with the Level A requirements of MIL-M-18058, and the following additional requirement.

5.1.1.1 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062 as specified for Type II, Method I.

5.1.2 Level C.

5.1.2.1 Engine. The fuel system, combustion chambers and valves, and when applicable, the cooling system shall be preserved in accordance with MIL-E-10062 as specified for Type II, Method I.

5.1.2.2 Repair parts and accessories. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts and accessories.

5.1.2.3 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.2.4 Other components. Other components of the track shifter, not specifically mentioned herein, shall be preserved in a manner to provide protection against deterioration and damage from the contractor to the initial destination.

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5.2 Packing and marking. Each complete track shifter, preserved as specified in 5.1 shall be packed and marked in accordance with MIL-M-18058. The degree of packing shall be Level A, Level B, or Commercial as specified (see 6.2).

6. NOTES

6.1 Intended use. The track shifter is intended for horizontal movement of track in either direction when relocating and realining railroad track.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of track shifter required (see 1.2).
(c) Time frame required for submission of the first-produced track shifter (see 3.2).
(d) Repair parts, maintenance tools, and accessories required (see 3.25).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the track shifter covered by this specification.

6.4 First-produced track shifter. Any changes or deviations of production track shifters from the approved first-produced track shifter during production will be subject to the approval of the contracting officer. Approval of a first-produced track shifter will not relieve the contractor of his obligation to furnish track shifters conforming to this specification.

6.5 Government serial numbers. The contracting officer should furnish the Government serial numbers for the track shifters upon award of the contract or order.

6.6 Recycled material. It is encouraged that recycled materials be used when practical as long as it meets the requirements of the specification (see 3.3).

6.7 Metric conversions. Metric conversion figures have been rounded to maintain implied tolerances.

Custodian:
Army - ME

Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE MIL-T-52289C(ME) Track Shifters, Railway, Gasoline- or Diesel-Engine-Driven

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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